

Finite-Field Chebyshev Dynamics: Ramified Functional Graphs, Zeta, and Koopman Jordan Data

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Abstract

For every prime power q and integer $d \geq 1$, we classify the complete functional graph of the non-normalized Chebyshev map T_d on $\mathbb{F}_q$. The graph is the inversion quotient of power maps on cyclic tori of orders $q-1$ and $q+1$, with their ramified intersection glued exactly. This gives every fixed and primitive count and the finite source zeta. Prime-support splitting further gives the full tail filtration, all image ranks, every zero Jordan-block size, and the full-function Koopman characteristic polynomial, without pretending that branch-folded trees are uniform.

1 Two cyclic covers and ramified gluing

Normalize the first-kind Dickson–Chebyshev family by

$$T_0(X) = 2, \quad T_1(X) = X, \quad T_d(X) = XT_{d-1}(X) - T_{d-2}(X).$$

Then $T_d(z + z^{-1}) = z^d + z^{-d}$ in every characteristic. In $\mathbb{F}_{q^2}^*$ put

$$G_- := \mathbb{F}_q^*, \quad G_+ := \ker N_{\mathbb{F}_{q^2}/\mathbb{F}_q}, \quad \eta(z) := z + z^{-1}.$$

The cover orders are $q-1, q+1$, and $|G_- \cap G_+| = \gcd(2, q-1)$. For $u \mid q-1, v \mid q+1$, define

$$\mathcal{Q}_q(u, v) = \frac{u + \gcd(2, u)}{2} + \frac{v + \gcd(2, v)}{2} - 1 - \mathbf{1}_{q \text{ odd}, 2 \mid u, 2 \mid v}. \quad (1)$$

This is the size of the two inversion-quotient subgroups after subtracting their common $+1$ branch and, when present twice, their common -1 branch.

Theorem 1 (complete labeled graph and all cycles). *Take disjoint functional graphs of $z \mapsto z^d$ on G_- and G_+ , identify $z \sim z^{-1}$ in each, and glue the two images of every element of $G_- \cap G_+$. The induced graph is conjugate through η to the complete functional graph of $T_d : \mathbb{F}_q \rightarrow \mathbb{F}_q$.*

For $D = d^n$ and $N \in \{q-1, q+1\}$ set

$$s_N(D) = \gcd(D-1, N) + \gcd(D+1, N) - \gcd(D-1, D+1, N),$$

$$i_N(D) = 1 + \mathbf{1}_{2 \mid N, D \text{ odd}}.$$

Then

$$F_n := |\text{Fix}(T_d^n)| = \sum_{N=q-1, q+1} \frac{s_N(D) + i_N(D)}{2} - 1 - \mathbf{1}_{q \text{ odd}, D \text{ odd}}. \quad (2)$$

Thus exact-period points and cycles are

$$E_m = \sum_{e \mid m} \mu(m/e) F_e, \quad C_m = E_m/m, \quad (3)$$

and

$$\zeta_{T_d}(t) = \exp \left(\sum_{n \geq 1} F_n \frac{t^n}{n} \right) = \prod_{m \geq 1} (1 - t^m)^{-C_m}. \quad (4)$$

Proof. The roots of $Z^2 - xZ + 1$ are inverse. If they are not in $\mathbb{F}_q$, Frobenius exchanges them, so $z^q = z^{-1}$ and they lie in G_+ . Hence η is surjective and has precisely the stated inversion fibres and glued intersection. The Chebyshev identity intertwines every directed edge.

Upstairs, a fixed lift obeys $z^D = z$ or $z^D = z^{-1}$. The union of these two cyclic kernels has size $s_N(D)$. Burnside adds its $i_N(D)$ inversion-fixed lifts and divides by two. The last two terms of (2) remove the branch values counted on both covers. Möbius inversion proves (3), and expanding $-\log(1 - t^m)$ proves (4). $\square$

The quotient statement is also the full tree theorem: regular inverse pairs fold together, while the one characteristic-two branch or two odd-characteristic branches retain their exceptional folded and glued trees. This construction follows the arbitrary-degree structural lineage of Qureshi–Panario [1]; Gassert treated the earlier prime-degree case [2]. Workspace ownership does not assert literature priority.

2 Tail, image, and Koopman Jordan atlases

Factor

$$q - 1 = a_- b_-, \quad q + 1 = a_+ b_+,$$

where $(a_{\pm}, d) = 1$ and every prime of $b_{\pm}$ divides d . The periodic population and the population of tail at most j are respectively

$$P = \mathcal{Q}_q(a_-, a_+), \quad H_j = \mathcal{Q}_q(a_- \gcd(d^j, b_-), a_+ \gcd(d^j, b_+)). \quad (5)$$

Thus exact tail zero has size P , exact tail $j \geq 1$ has size $H_j - H_{j-1}$, and the height is the least h with $b_- \mid d^h$ and $b_+ \mid d^h$. Equation (5), together with the labeled quotient theorem, retains each local tree rather than inferring a false uniform tree after ramification.

The image of the j th power has cover orders

$$m_{\pm}(j) = \frac{q \pm 1}{\gcd(d^j, q \pm 1)},$$

so

$$R_j := |\operatorname{im} T_d^j| = \mathcal{Q}_q(m_-(j), m_+(j)). \quad (6)$$

These numbers stabilize at P .

Let $U : \mathbb{C}^{\mathbb{F}_q} \rightarrow \mathbb{C}^{\mathbb{F}_q}$ be composition, $(Uf)(x) = f(T_d(x))$. Distinct rows of U^j correspond to distinct image values; hence $\operatorname{rank} U^j = R_j$. Second differences give the number of zero Jordan blocks of exact size j ,

$$Z_j = R_{j-1} - 2R_j + R_{j+1}. \quad (7)$$

The recurrent quotient is the cycle permutation from (3), and therefore

$$\det(\lambda I - U) = \lambda^{q-P} \prod_{m \geq 1} (\lambda^m - 1)^{C_m}. \quad (8)$$

The identity face $d = 1$ is included. The separate $d = 0$ face has $T_0 = 2$ (zero in characteristic two): one fixed point, $q - 1$ points of tail one, zeta $(1 - t)^{-1}$, ranks $q, 1, 1, \dots$, and a diagonalizable composition operator with eigenvalues $1, 0^{q-1}$.

References

- [1] C. Qureshi and D. Panario, *The graph structure of Chebyshev polynomials over finite fields and applications*, *Designs, Codes and Cryptography* **87** (2019), 393–416, [doi:10.1007/s10623-018-0545-7](https://doi.org/10.1007/s10623-018-0545-7).
- [2] T. A. Gassert, *Chebyshev action on finite fields*, *Discrete Mathematics* **315–316** (2014), 83–94, [doi:10.1016/j.disc.2013.10.014](https://doi.org/10.1016/j.disc.2013.10.014).